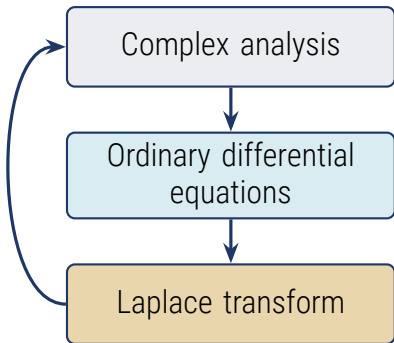




MAT290: Advanced Engineering Mathematics

Lecture 14: Why Laplace Transform?

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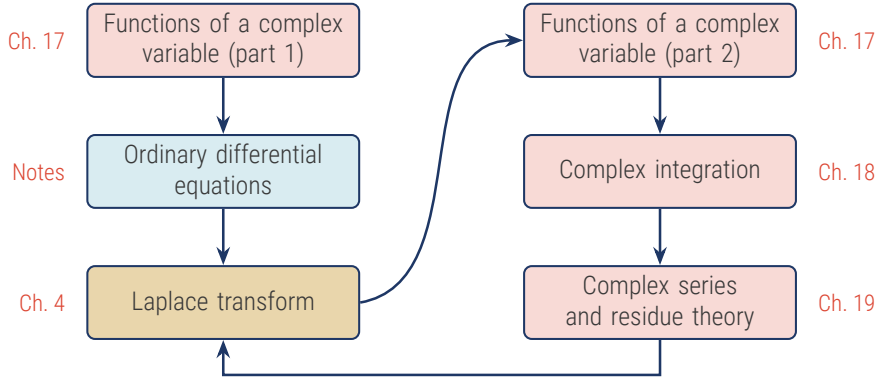
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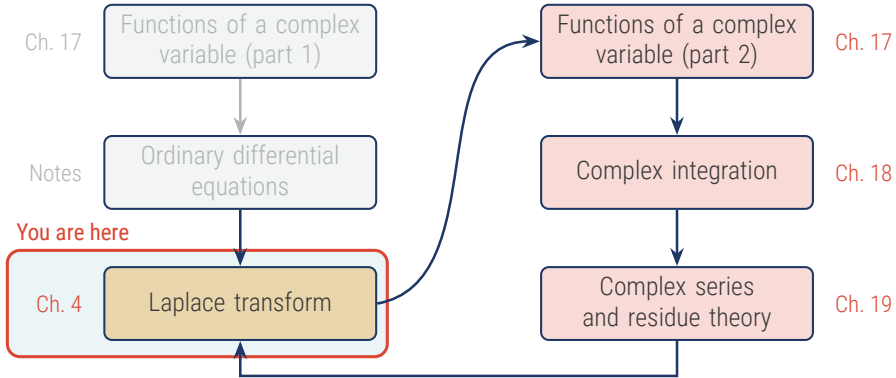


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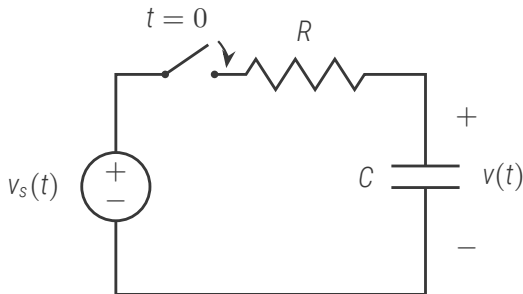
Our Progression in This Course



Our Progression in This Course



Why the Laplace Transform?



$$\begin{aligned}v_s(t) &= 5 \cos(\omega t), \\ \omega &= 2\pi \times 10^6 \text{ rad/s}, \\ R &= 250 \text{ k}\Omega, \\ C &= 6.5 \text{ }\mu\text{F}.\end{aligned}$$

- The switch closes at $t = 0$. We want the capacitor voltage $v(t)$.
- We already have two tools for analyzing this circuit:
 - Phasor analysis (from circuit courses , e.g., ECE212)
 - Time-domain differential equations (from this course)

- Represent a sinusoidal signal by its amplitude and phase.

$$A \cos(\omega t + \theta) \longleftrightarrow \bar{V} = Ae^{j\theta} = A\angle\theta.$$

Time-domain relations

Resistor: $v_R(t) = Ri(t),$

Capacitor: $i(t) = C \frac{dv_C}{dt}.$

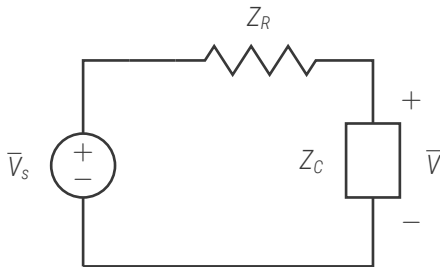
Phasor-domain relations

$$\bar{V}_R = Z_R \bar{I}, \quad Z_R = R,$$

$$\bar{V}_C = Z_C \bar{I}, \quad Z_C = \frac{1}{j\omega C}.$$

- At a fixed frequency, differentiation becomes multiplication by $j\omega$.

The RC Circuit in the Phasor Domain



$$Z_R \frac{\bar{V}}{Z_C} + \bar{V} = \bar{V}_s \Rightarrow \bar{V} = \frac{\bar{V}_s}{1 + Z_R/Z_C} = \frac{\bar{V}_s}{1 + j\omega RC}.$$

Time-domain ODE
 $v + RC\dot{v} = v_s(t)$

Transform

Phasor-domain algebra
 $\bar{V} = \bar{V}_s / (1 + j\omega RC)$

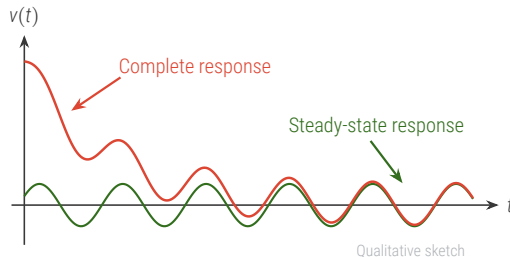
Invert

Steady-state response
 $v_{ss}(t) = \text{Re}\{\bar{V}e^{j\omega t}\}$

- The basic phasor method treats **sinusoidal sources** at a fixed frequency.
- It gives the **steady-state response**, without the switching transient.

$$v_{ss}(t) = A \cos(\omega t - \phi),$$
$$A = \frac{5}{\sqrt{1 + (\omega RC)^2}}, \quad \phi = \arctan(\omega RC).$$

The initial capacitor voltage determines an additional transient.



We stay in the time domain and solve

$$RC \frac{dv}{dt} + v = 5 \cos(\omega t).$$

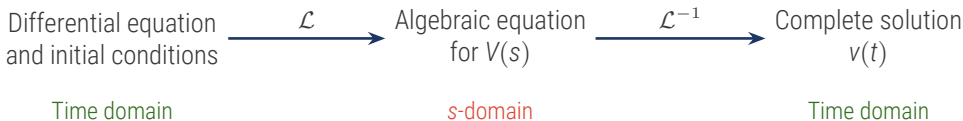
1. Find the complementary solution: $v_c(t) = Ke^{-t/(RC)}$.
2. Find a particular solution: $v_p(t) = A \cos(\omega t - \phi)$.
3. Combine them and use the initial condition to determine K .

$$v(t) = \underbrace{A \cos(\omega t - \phi)}_{\text{steady-state response}} + \underbrace{Ke^{-t/(RC)}}_{\text{transient response}}.$$

$$K = v(0) - A \cos \phi.$$

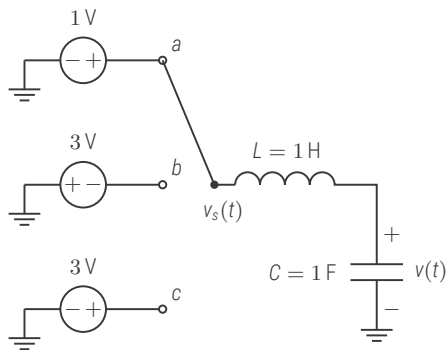
- These methods provide both transient and steady-state responses.
- Finding a closed-form solution can be difficult.
 - The methods learned so far apply to particular classes of ODEs, such as separable equations and linear equations with constant coefficients.
- Even when a method applies, the calculation can be tedious.
- Finding a particular solution is harder for general inputs.
 - What if the forcing function is discontinuous?

- Transform the ODE from the time domain to the s-domain.
- Perform algebraic manipulations, then transform back.

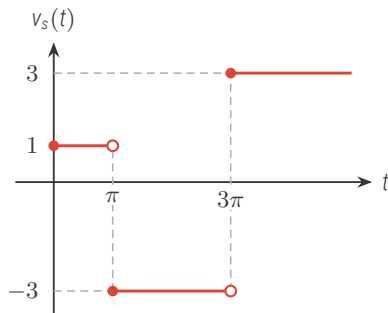


- This incorporates the transient and steady-state terms together.
- It is especially useful for linear ODEs with switching or discontinuous inputs.

A Preview: An ODE with a Discontinuous Input



$$\ddot{v} + v = v_s(t), \quad v(0) = \dot{v}(0) = 0,$$



$$v_s(t) = \begin{cases} 1, & 0 \leq t < \pi, \\ -3, & \pi \leq t < 3\pi, \\ 3, & t \geq 3\pi. \end{cases}$$

A Preview: The Magic of Laplace Transform

- Write $V(s) = \mathcal{L}\{v(t)\}$. The transformed ODE is

$$s^2 V(s) - \underbrace{s v(0)}_0 - \underbrace{\dot{v}(0)}_0 + V(s) = \mathcal{L}\{v_s(t)\}.$$

- The initial conditions are already included in $V(s)$.
- For the switched input, with Laplace Transform, you can derive

$$\mathcal{L}\{v_s(t)\} = \frac{1 - 4e^{-\pi s} + 6e^{-3\pi s}}{s}.$$

- Solving the algebraic equation in the s -domain gives

$$V(s) = \frac{1}{s^2 + 1} \left(\frac{1 - 4e^{-\pi s} + 6e^{-3\pi s}}{s} \right).$$

- Applying the inverse Laplace transform gives

$$v(t) = \mathcal{L}^{-1}\{V(s)\} = \begin{cases} 1 - \cos t, & 0 \leq t < \pi, \\ -3 - 5 \cos t, & \pi \leq t < 3\pi, \\ 3 + \cos t, & t \geq 3\pi. \end{cases}$$

Definition of Laplace Transform